

Kiseki no Mock
December 24, 2024

H. 225

Ans: D

3. There are three towns: Astana, Almaty and Aktobe.

The distance from Astana to Almaty is the same as the distance from Astana to Aktobe.

The distance from Astana to Aktobe is 1300 km.

Aktobe is on west from Astana and Almaty is on bearing of 150° from Astana.

What is the distance from Almaty to Aktobe in kilometres?

- A. $1300\sqrt{3}$
- B. $650\sqrt{3}$
- C. $1300\sqrt{2}$
- D. 1300
- E. 2600

Ans: A

4. $x^2 - bx + 5 = 0$, where b is positive constant, has two solutions that differ by 4.

What is the value of b ?

- A. 4
- B. -6
- C. 6
- D. -4
- E. 10
- F. 8

Ans: C

5. Here is the formula for the period of oscillation of a charged particle in a magnetron:

$$T = 2\pi \sqrt{\frac{m}{qB^2} \left(\frac{k}{qE} - r \right)}$$

Rearrange the formula to make r the subject.

A. $r = \frac{k}{qE} + \frac{T^2 q B^2}{4\pi^2 m}$

B. $r = \frac{T^2 q B^2}{4\pi^2 m} - \frac{k}{qE}$

C. $r = \frac{k}{qE} - \frac{T^2 q B^2}{4\pi^2 m}$

D. $r = \frac{k}{qE} - \frac{4\pi^2 m}{T^2 q B^2}$

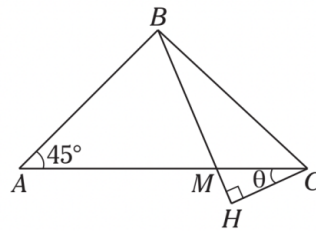
E. $r = \frac{k}{qE} - \frac{T q B^2}{2\pi m}$

F. $r = \frac{k}{qE} - \frac{T^2 q^2 B^2}{4\pi^2 m}$

G. $r = \frac{k}{qE} - \frac{T^2 q B^2}{2\pi m}$

Ans: C

6. Find $\tan \theta$, if $AB = 8\sqrt{2}$ and $AM = 11$



- A. $\frac{3}{8}$
- B. $\frac{5}{8}$
- C. $\frac{1}{2}$
- D. $\frac{1}{5}$
- E. $\frac{8}{3}$

Ans: A

7. Find an equation for the line perpendicular to $4x - 5y = 20$ and sharing the same y-intercept

- A. $5y - 4x = 80$
- B. $-5x + 4y = -20$
- C. $5x - 4y = 80$
- D. $4x + 5y = 16$
- E. $5x + 4y = -16$

Ans: E

8. A draft person makes a scale drawing of 100 meter $\times$ 30 meter building, where 1 centimeter represents 2.5 meters. How many centimeters are in the smaller dimension of the building?

- A. 18 centimeters
- B. 12 centimeters
- C. 480 centimeters
- D. 8 centimeters
- E. 40 centimeters

Ans: B

9. Find the smallest positive integer x so that the fraction $\frac{1}{10+x}$ has a finite decimal

- A. 5
- B. 2
- C. 10
- D. 6
- E. 15

Ans: D

10. Which is the largest prime factor of $4^6 - 1$?

- A. 13
- B. 17
- C. 5
- D. 3
- E. 23
- F. 31

Ans: A

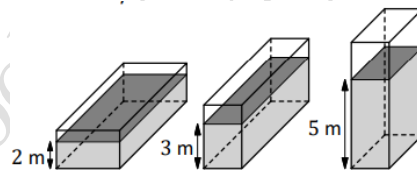
11. Let x and y be positive integers such that $\frac{1}{x} + \frac{1}{y} = \frac{1}{20}$. What is the maximum possible value of y ?

- A. 40
- B. 60
- C. 240
- D. 420
- E. 480

Ans: D

12. A cuboid-shaped container that is not filled completely contains 120 m^3 of water. The depth of the water is either 2 m or 3 m or 5 m, depending on which side the container is actually standing on (drawings not to scale).

How big is the volume of the container?



- A. 160 m^3
- B. 240 m^3
- C. 180 m^3
- D. 220 m^3
- E. 200 m^3

Ans: B

13. The expression $\frac{7n+12}{2n+3}$ takes integer values for certain integer values of n .
What is the sum of all such integer values of the expression?

A. 4
B. 8
C. 10
D. 12
E. 14

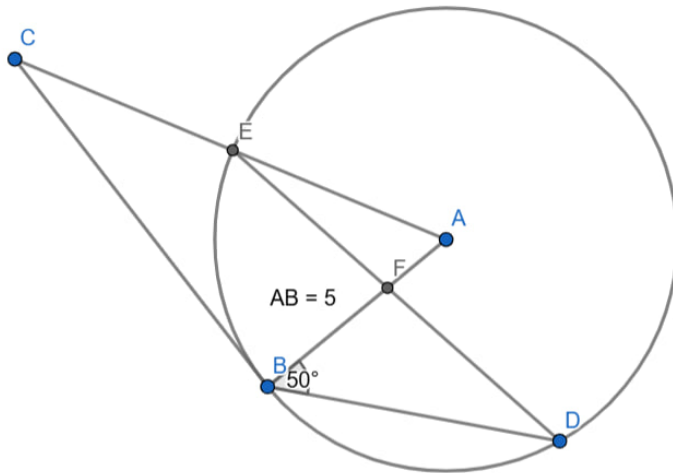
Ans: E

14. Which of the answers below is equal to $0.1(7)$?

A. $\frac{32}{180}$
B. $\frac{17}{99}$
C. $\frac{17}{90}$
D. $\frac{16}{99}$
E. $\frac{7}{45}$
F. $\frac{8}{46}$

Ans: A

15. The line BC is a tangent at the point B to a circle with centre A .
 $AC = 10$
Find angle FEA .



- A. 10°
- B. 15°
- C. 17.5°
- D. 20°
- E. 25°
- F. 30°
- G. 40°

Ans: D

16. If x , y , and z are positive integers and $2^x \cdot 3^y \cdot 5^z = 54000$, what is the value of $x + y + z$?

- A. 8
- B. 9
- C. 10
- D. 11
- E. 12

Ans: C

17. In my desk, the number of pencils and pens was in the ratio 4 : 5. I took out a pen and replaced it with a pencil and now the ratio is 7 : 8. What is the total number of pencils and pens in my desk?

- A. 19
- B. 20
- C. 45
- D. 12
- E. 31

Ans: C

18. 60% of a number is equal to one-thirds of a second number.

The whole number ratio of the first number to the second number in its lowest terms is $x : y$

What is the value of $x - y$?

- A. 1
- B. -2
- C. 2
- D. -1
- E. 4
- F. -4
- G. 0.8
- H. -0.8

Ans: F

19. Given that

$$125^{2x-2} \cdot \frac{1}{25^{2x+1}} = 625^x$$

What is the value of x ?

- A. -4
- B. 4
- C. -2
- D. 0
- E. 8

Ans: A

20. Solve the inequality:

$$\frac{2}{5}(x-3) - \frac{1}{4}(x+7) < 3$$

- A. $x > \frac{119}{3}$
- B. $x < \frac{109}{3}$
- C. $x < \frac{119}{3}$
- D. $x > \frac{109}{3}$
- E. $x < \frac{119}{4}$
- F. $x > \frac{119}{3}$

Ans: C

21. The point $(-2, 6)$ is translated to the point $(4, -3)$ by two successive translations.

The first translation is by the vector

$$\begin{pmatrix} 2p \\ -5p \end{pmatrix}$$

The second translation is by the vector

$$\begin{pmatrix} q \\ -3q \end{pmatrix}$$

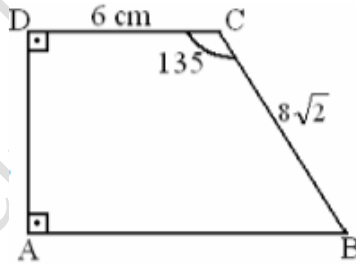
What is the value of $p + q$?

- A. 15
- B. 6
- C. 0
- D. -3
- E. -9

Ans: D

22. ABCD is a right trapezoid where

$$m(\angle C) = 135^\circ, \quad |DC| = 6 \text{ cm}, \quad |BC| = 8\sqrt{2} \text{ cm}.$$



Find the area $A(ABCD)$.

- A. 48
- B. $48\sqrt{2}$
- C. 80
- D. 120
- E. 160

Ans: C

23. The vertices of a rectangle have coordinates:

$$P(3, 4), \quad Q(3, 9), \quad R(8, 9), \quad S(8, 4).$$

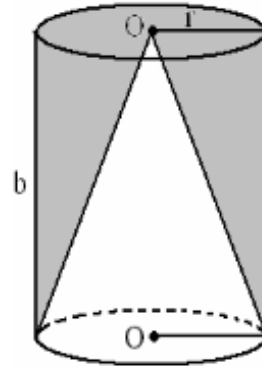
$PQRS$ is transformed by a clockwise rotation of 90° about P followed by a reflection in the x-axis.

What are the coordinates of the final position of R ?

- A. $(-9, -8)$
- B. $(8, 1)$
- C. $(4, -1)$
- D. $(6, -10)$
- E. $(-6, -11)$

Ans: B

24. The volume of the cylinder is V . Find the volume of the region between cylinder and cone in terms of V .



- A. V
- B. $\frac{3V}{4}$
- C. $\frac{2V}{3}$
- D. $\frac{V}{3}$
- E. $\frac{V}{2}$

Ans: C

25. The equation of a line is given as:

$$qx + py = 5.$$

The line passes through the points $(2, 3)$ and $(4, 7)$.

Find the value of $q + p$.

- A. 5
- B. 7
- C. -3
- D. 10
- E. 0

Ans: A

26. Solve the inequality:

$$\frac{1}{2}(x - 2) - \frac{2}{3}(3 - x) < x - 4$$

- A. $x < -6$
- B. $x < \frac{7}{3}$
- C. $x > -5$
- D. $x > \frac{6}{7}$
- E. $x > 12$

Ans: A

27. The quadratic expression:

$$x^2 - 12x + 35$$

has been transformed to the form:

$$p(x + q)^2 - r$$

Find the value of $3p + q$.

- A. -7
- B. 5
- C. -2
- D. 10
- E. -3

Ans: E

28. For what values of a does the equation:

$$(1 - a)x^2 + (a + 1)x - 2 = 0$$

have two distinct real roots?

- A. $a < 0, a > 3$
- B. $a < 0$
- C. $a \neq 3$
- D. $a \neq 3, a \neq 0$
- E. $0 < a < 3$

Ans: D

29. There is a parallelogram $ABCD$ oriented clockwise. $AB = \vec{m}$, $AD = \vec{n}$. There is a point X such that $\frac{AX}{XC} = \frac{1}{5}$.

Express the vector $\vec{XD}$ in terms of $\vec{m}$ and $\vec{n}$.

- A. $\frac{1}{5}\vec{m} - \frac{5}{6}\vec{n}$
- B. $\frac{1}{6}\vec{m} - \frac{5}{6}\vec{n}$
- C. $\frac{1}{6}(\vec{m} + \vec{n})$
- D. $\frac{1}{5}(4\vec{n} - \vec{m})$
- E. $\frac{1}{6}(5\vec{n} - \vec{m})$
- F. $\vec{m} + \frac{5}{6}\vec{n}$
- G. $\frac{1}{6}(\vec{m} + \vec{n})$
- H. $\frac{5}{6}(\vec{m} + \vec{n})$

Ans: E

30. P is directly proportional to Q^2 .

When $P = 2$, $Q = 4$.

Q is inversely proportional to R .

When $Q = 2$, $R = 5$.

What is P in terms of R ?

A. $P = \frac{5}{R}$

B. $P = \frac{5}{4R}$

C. $P = \frac{1}{800R^2}$

D. $P = \frac{5}{4R^2}$

E. $P = \frac{800}{R^2}$

F. $P = \frac{R^2}{50}$

G. $P = \frac{25R^2}{2}$

H. $P = \frac{25}{2R^2}$

Ans: H

Critical Thinking

1. Abigail counted to 2400 by 6's beginning with 6. Sebastian counted to 2400 by 4's starting with 4. How many of the numbers counted by Abigail were also counted by Sebastian?

A. 180
B. 230
C. 120
D. 350
E. 400
F. 200

Ans: F

2. A tech company claims that implementing AI in customer service will drastically improve customer satisfaction by reducing wait times and providing accurate responses. The company presents data showing that AI systems handle inquiries faster than human agents. However, critics note that AI lacks the empathy and understanding required to handle complex or emotionally charged situations. The company argues that speed and efficiency are the most important factors for improving customer satisfaction.

Which **one** of the following best expresses the flaw in the above argument?

A. The argument does not consider the cost of implementing AI systems.
B. The argument fails to address situations where human judgment is necessary.
C. The argument does not provide examples of industries where AI has succeeded.
D. The argument assumes that all customer inquiries are straightforward.
E. The argument assumes that speed is more important to customers than empathy

Ans: E

3. Recent studies reveal that the consumption of sugary beverages significantly increases the risk of developing chronic illnesses like diabetes and heart disease. Efforts to educate the public on healthier alternatives have shown limited success, mainly due to aggressive marketing by beverage companies. To address this, governments should impose higher taxes on sugary drinks to deter consumption and fund public health initiatives.

What is the main conclusion of the passage below?

- A. Public education efforts alone cannot reduce sugary drink consumption.
- B. Higher taxes on sugary drinks can effectively reduce consumption and support public health.
- C. Sugary beverages are the leading cause of chronic illnesses.
- D. Aggressive marketing is the primary reason for increased sugary drink consumption.
- E. Governments should ban sugary beverages entirely to protect public health.

Ans: B

4. Two standard dice are placed on a table, with one on top of the other, so that only nine of the faces of the dice may be seen. The touching faces have the same number on them. The sum of the numbers on the visible face is 33

What is the number on the touching faces?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6

Ans: B

5. Critics of standardized testing argue that it places undue stress on students and fails to capture the complexity of their abilities. In particular, they point out that these tests emphasize rote memorization over critical thinking and problem-solving skills. As a result, a group of educators has proposed replacing standardized testing with a more holistic approach, such as portfolio assessments and project-based evaluations.

Which one of the following is an underlying assumption of the argument above?

- A. Stress from standardized testing is the primary barrier to student success.
- B. Standardized tests do not evaluate non-academic abilities.
- C. Holistic assessment methods will provide a more accurate picture of students' skills.
- D. Educators universally agree on eliminating standardized testing.
- E. Portfolio assessments are already proven to be superior to standardized tests.

Ans: C

6. The table below shows 'audience numbers' and 'time per visit' for a number of popular social networking sites. A comparison of figures for the month of February is shown for two consecutive years.

Site	February 2008		February 2007	
	Audience (000's)	Time per visit (hr:min:sec)	Audience (000's)	Time per visit (hr:min:sec)
myspace.com	55,419	2:12:19	53,362	2:00:24
Facebook	20,043	1:06:43	9,923	1:04:54
LinkedIn	7,392	0:10:31	1,990	0:05:10
Flixster	2,619	0:07:38	1,591	0:12:28
Reunion.com	4,323	0:04:34	4,348	0:04:49
Meetup.com	1,940	0:12:16	1,215	0:07:07
Last.fm	1,938	0:06:43	1,508	0:02:24
myYearbook	1,738	0:07:02	2,368	0:09:04

How many sites had an increase in both 'audience numbers' and 'time per visit' from 2007 to 2008?

- A. 5
- B. 6
- C. 2
- D. 3

E. 1

F. 4

Ans: A

7. Scientists have long debated the role of artificial light pollution on ecosystems. Recent research highlights how excessive artificial light disrupts the natural behaviors of nocturnal animals, such as hunting and mating patterns. It also shows that light pollution alters the growth cycles of plants by affecting their exposure to natural light. To mitigate these harmful effects, cities should adopt policies that regulate and reduce unnecessary outdoor lighting, especially in sensitive ecological areas.

Which one of the following best expresses the main conclusion of the above argument?

- A. Cities should implement policies to reduce unnecessary outdoor lighting in ecological areas.
- B. Artificial light pollution disrupts the behavior of nocturnal animals.
- C. Plants are affected by artificial light, which alters their growth cycles.
- D. Light pollution has been widely studied and debated by scientists.
- E. Outdoor lighting is unnecessary in most urban areas.

Ans: A

8. Five friends are dealt two cards each from a set of twelve cards. The cards are numbered 1 to 12 inclusive. In turn, the friends declare the sum of the values of their two cards, Linus scores 4, Lewis scores 11, Pam scores 16, Willy scores 19 and Clint scores 20.

Which of the following statements is true?

- A. Linus has card 2
- B. Lewis has card 3
- C. Pam has card 5
- D. Willy has card 7
- E. Clint has card 11

Ans: C

9. Jodi has a set of ten fridge magnets showing the integers from 0 to 9 inclusive.

In how many different ways can she split the set into five pairs so that the sum of each pair is a multiple of 5?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Ans: D

10. A recent study claims that regular exercise reduces the risk of heart disease by improving cardiovascular health and maintaining a healthy weight. However, some individuals argue that the time commitment and physical strain discourage people from exercising regularly. To promote widespread physical activity, it is suggested that governments invest in public fitness programs and offer subsidies for gym memberships.

Which of the following strengthens the conclusion that governments should invest in promoting exercise?

- A. Studies show that people with gym memberships are not significantly healthier than those without.
- B. Countries with public fitness initiatives have seen a reduction in heart disease rates over the past decade.
- C. Many individuals prefer non-physical methods, such as dieting, to maintain a healthy lifestyle.
- D. Public fitness programs often require large financial investments without guaranteed participation.
- E. Exercise is already widely promoted by non-government organizations.

Ans: B

11. Alpha, Beta and Gamma are the three political parties in the Parliament.

There are 360 seats in the Parliament. Before last week's General Election, each party held 120 seats.

In last week's election:

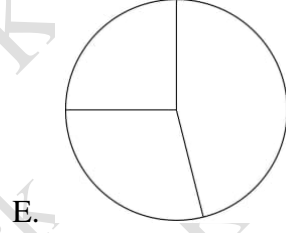
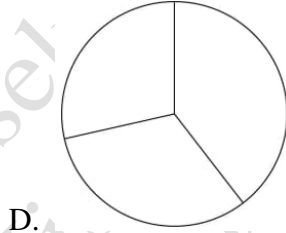
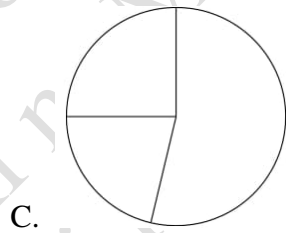
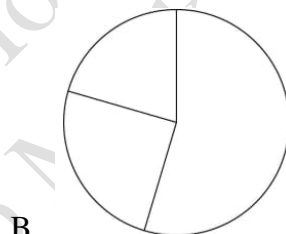
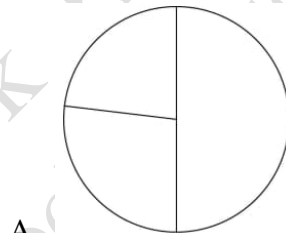
There was no change in 216 seats.

Alpha gained 15 seats from Beta and 23 seats from Gamma.

Beta gained 31 seats from Alpha and 41 seats from Gamma.

Gamma gained 24 seats from Alpha and 10 seats from Beta.

Which pie chart, suitably labelled, shows the current state of the parties in the Parliament?



Ans: E

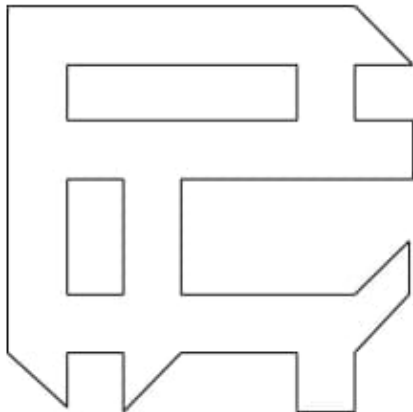
12. Food manufacturers should reduce sugar levels in processed foods. This is because there are too many overweight and obese people. Since people are unable or unwilling to make sensible decisions for themselves, manufacturers must take responsibility for rectifying a problem to which they contribute.

Which one of the following illustrates the principle used in the above argument?

- A. Cigarette packets should have plain packaging and display government health warnings.
- B. In order to reduce sugar intake, foods with high sugar content should be taxed more heavily
- C. Power tools should have clear instructions in order to reduce the number of accidents caused by incorrect use.
- D. To reduce the number of deaths caused by excessive speed, car manufacturers should incorporate speed limiters in vehicles.
- E. Labelling on food should be improved so that consumers are clear about the level of sugar in a product.

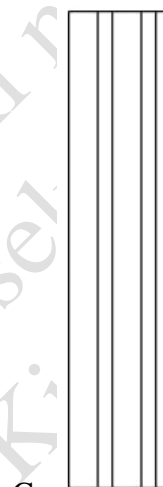
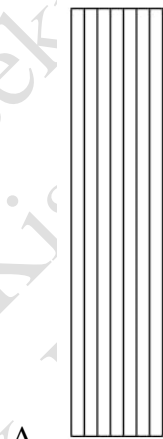
Ans: D

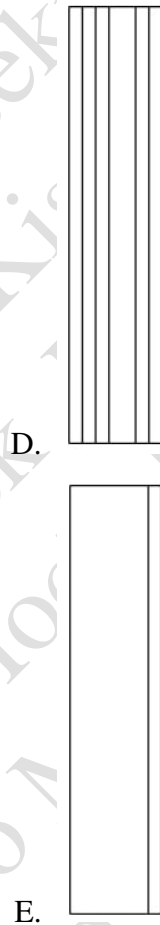
13. This is a view of a cross section of part of an aluminium window frame.



The cross section is uniform throughout its length.

Which one of the following is NOT a possible side view of this part of the aluminium window frame?





Ans: C

14. David normally takes 45 minutes to walk at a steady speed in a straight line from his house, past his friend Barry's house and then past a shop to reach his college. Today, however, he stopped at Barry's house on the way and had to wait 3 minutes for him to be ready to leave. They stopped at the shop to buy their lunch, which took another 3 minutes.

It normally takes David 20 minutes to walk from his home directly to the shop and Barry normally takes 32 minutes to walk directly from his home to the college and they both walk at the same steady speed.

How long does it take them to walk from Barry's house to the shop?

- A. 7 minutes
- B. 1 minute
- C. 13 minutes

- D. 4 minutes
- E. 10 minutes

Ans: A

15. What is the secret of success? The question is as old as philosophy itself. The Greeks argued that it boiled down to the possession and exercise of certain character traits, such as generosity and bravery. But new research emerging from social psychology suggests you have to be selfconfident to be successful. The study, involving over 200 successful people from all walks of life, comprised detailed interviews and the observation of mental and physical problem solving. The research found that all of the participants had a strong belief in their own abilities and persevered in the face of challenges because they thought they could do it. If you want to be successful, then, you must have faith in your own abilities.

Which one of the following is the best statement of the flaw in the above argument?

- A. It assumes that self-confidence is responsible for success, and not vice versa.
- B. It relies on different senses of the expression 'self-confidence'
- C. It ignores the significance of other factors, such as luck.
- D. It assumes that everyone can become self-confident.
- E. It justifies a claim on the basis of how things were in the past.

Ans: A

16. There are 10 students in a class and they like visiting each other. A visit means that one student visits the home of another student. We know that each student in the class has visited the same number of students. No two students in the class have both visited each other.

What is the maximum number of students that a student in this class could have visited?

- A. 2
- B. 4
- C. 5
- D. 8
- E. 9

Ans: B

17. The presence of a mind requires the presence of self-consciousness. Machines do not have self-consciousness, therefore machines do not have minds.

Which one of the following most closely parallels the reasoning used in the above argument?

- A. The morally best option is that which maximises benefit. Larger charities are able to provide the most benefit from donations, therefore it is morally best to choose larger charities to donate to.
- B. There is no obligation to keep promises to someone after they have died, since the virtue of keeping promises is to promote the promisee's wellbeing, and their capacity for wellbeing ceases upon death.
- C. To count as an emotion, a mental state needs to have representational content. Depression does not have representational content, therefore depression is not an emotion.
- D. Trust and reliance are not the same thing, because trust is a mental attitude, whereas reliance is not.
- E. Only true beliefs can be knowledge. My belief that the world is spherical is true, therefore I know that the world is spherical.

Ans: C

18. In a survey of cats' eating habits, 10% of the cats tested showed a clear preference for Moggiemeat, 20% preferred Fuss-a-Puss and 40% preferred Yummy Paws. An additional 6 cats did not seem to care at all which of the three foods they ate, while the remaining 27 cats in the survey would not touch any of these foods.

How many cats were involved in the survey?

- A. 33
- B. 63
- C. 90
- D. 103
- E. 110

Ans: E

19. 'Internet addiction' is about to be classified as a recognised psychiatric disorder. Internet addiction is characterised by a number of signs: excessive use of the internet; anger or depression if access is lost; isolation from friends and family; and, most significantly, poor educational achievement. This should make us cautious about incorporating computer-based learning into all aspects of education. Educationalists should consider the long-term problems associated with extensive internet use as well as its immediate attractions.

Which one of the following is an assumption of the above argument?

- A. Students are unable to regulate their internet use
- B. It is a bad idea to incorporate computer-based learning into education
- C. Classifying internet addiction as a recognised psychiatric disorder will help lead to a cure.
- D. All computer-based learning includes internet use.
- E. Excessive internet use is the only cause of poor educational achievement.

Ans: D

20. Next week the European Parliament will vote on whether to ban trawling, a method of fishing that involves pulling large fishing nets behind boats. The fishing industry is opposed to a ban and has argued strongly against it. Trawling is not as damaging to ecosystems as environmentalists claim. Trawling the seabed doesn't always create 'dead zones' in the ocean. It can boost fish numbers, since the species that are more resistant to the effects of trawling can proliferate. So trawling on flat sandy beds in shallow areas can benefit marine life.

Which one of the following, if true, most weakens the above argument?

- A. Trawling creates new habitats by making parts of the sea more habitable
- B. Trawling in deep waters is expensive and an ineffective use of fishing resources.
- C. Trawling nets have a destructive impact on many endangered species.
- D. Trawling in shallow waters can damage the bottom of fishermen's boats.
- E. Trawling in shallow waters can alter the ecology in beneficial ways.

Ans: C

21. Childhood obesity is still a problem. This is often blamed on the fact that children have easy access to a range of unhealthy foods and increasingly have more money available to them to spend on fattening and sugar-filled foods. A solution seems obvious—raise the prices of all these foods so that parents will have more control over what their children eat and will find it more economical to buy healthy foods. Unfortunately, this is unlikely to work as the problem is also caused by the fact that unhealthy foods tend to be more convenient and that is what many parents consider when deciding what foods to provide. A wider range of ready-to-eat healthy food options would be a good start to solving the problem of child obesity.

Which one of the following best expresses the main conclusion of the above argument?

- A. Parents should have more control over the food that their children eat.
- B. The price of unhealthy foods should be raised.
- C. There should be more "ready-to-eat" healthy meals available.
- D. Children have too much access to fattening and sugar-filled foods.
- E. Child obesity is still a problem.

Ans: C

22. In recent years, wearable health technology has become increasingly popular. These devices can track vital statistics such as heart rate, sleep patterns, and physical activity levels. Although they provide users with valuable insights into their health, many experts warn that relying solely on these gadgets could lead to overdiagnosis and unnecessary anxiety. To maximize their benefits, wearable technology should be used as a supplement to professional medical advice, not a replacement.

What is the main conclusion of the passage below?

- A. Wearable health technology has become increasingly popular in recent years.
- B. Overreliance on wearable health devices can lead to overdiagnosis and anxiety.
- C. Professional medical advice is more reliable than wearable health technology.
- D. Wearable health devices should complement, not replace, professional medical guidance.
- E. Tracking vital statistics is an essential feature of wearable health technology.

Ans: D

23. A café owner makes 50 mini pizzas and 50 flapjacks for sale during the day. The pizzas are sold for \$2.00 each and the flapjacks are sold for \$1.00 each. However, by the end of lunchtime, there are 10 mini pizzas still unsold and 15 flapjacks unsold, so the price of each is halved for the afternoon. At the end of the day, there are still 2 mini pizzas and 1 flapjack left, which are given away free to the last customer.

If the original 50 mini pizzas and 50 flapjacks had a combined production cost of \$30.00, how much profit is made on the sale of the mini pizzas and flapjacks for the day?

- A. \$100.00
- B. \$102.50
- C. \$115.00
- D. \$120.00
- E. \$130.00

Ans: A

24. The UK Department of Education is encouraging pupils aged 14 to 18 to learn poetry by heart in order to recite it at a new national competition. This government initiative is designed to boost the confidence of young people and increase their awareness of poetry. By taking part in this initiative and by reciting poetry, young people will improve their appreciation of poetry.

Which one of the following identifies a flaw in the above argument?

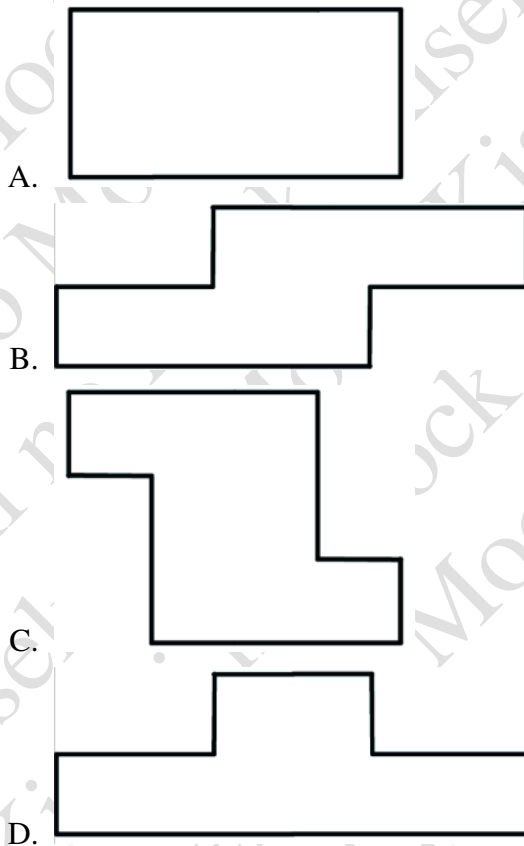
- A. Young people are already aware of poetry.
- B. You don't have to recite poetry to appreciate it.
- C. Awareness is not the same as appreciation.
- D. The young people who are willing to recite poetry are already confident.
- E. Learning something in order to recite it is not the same as being creative yourself.

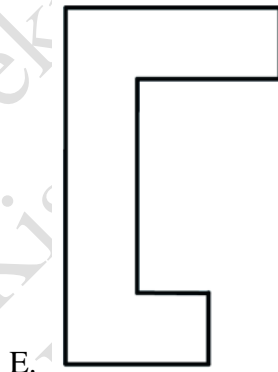
Ans: C

25. Marina purchased a pair of L-shaped tables thinking that they would add an unusual touch to her conference room.



Which one of the following arrangements could not be made using the two L-shaped tables?





Ans: D

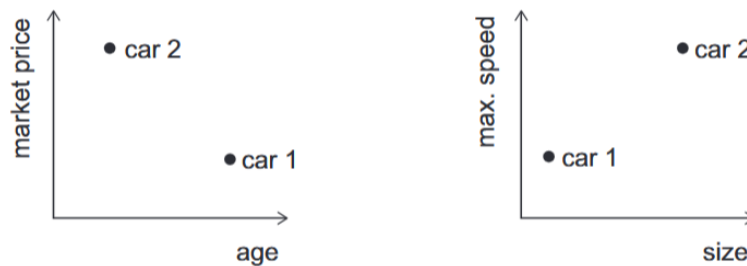
26. Drivers found to be committing offences such as speeding are now offered the choice between paying the police for a re-education course or taking the traditional penalty of paying a fine and receiving penalty points on their licence. It is no surprise that the number of motorists opting for re-education courses has tripled in five years, but the high cost of such courses has led to accusations that police forces might use the new system for their own benefit. They could easily set up speed traps, catch a number of motorists slightly over the limit, and generate easy income for themselves. Re-education courses are a good idea in theory, but open to abuse by cash-starved police forces.

Which one of the following is an underlying assumption of the above argument?

- A. The cost of running re-education courses is less than the combined contributions of participants.
- B. Re-education is more effective than punishment in preventing further motoring offences.
- C. Police forces are more interested in generating income than in preventing motoring offences.
- D. The number of driving offences committed in the last five years has tripled.
- E. Fear of losing a driving licence is greater than fear of financial hardship.

Ans: A

27. Peter has drawn two diagrams showing information about the two cars owned by his parents:



Which one of the following statements is?

- A. The more expensive car is newer.
- B. The cheaper car is bigger.
- C. The older car is slower.
- D. The newer car is bigger.
- E. The slower car is smaller.

Ans: B

28. A school party consisting of 4 teachers and 20 students is visiting the theatre. Of the students, 8 are aged 15 years and 12 are aged 16 years. The following table shows the entry prices to the theatre.

category	price
Adult (16 years and over)	\$30
Child (5–15 years)	\$20
Under 5 years	free
Family ticket (2 adults and up to 2 children under 16 years)	\$90
Group ticket (8 people including at least 2 children under 16 years)	\$175
Supergroup ticket (any 8 people)	\$200

What is the least amount that the school party must pay for entry to the theatre?

- A. \$525
- B. \$530
- C. \$545
- D. \$550
- E. \$560

Ans: A

29. It has emerged that prisoners have helped to maintain one of Britain's busiest railway lines. Inmates from an open prison have worked night-shifts on the main line from London to Glasgow. This is the same line where a train left the tracks earlier this year, killing one person and injuring several others. It is all very well to have offenders being rehabilitated through work, but what appears to be happening here is that rail contractors are cynically using prisoners as cheap labour. This is unacceptable, not just for safety reasons – important as they are – but because ultimately it is an infringement of workers' rights. If these prisoners are being paid less, or treated differently from regular workers, the practice must stop once and for all.

Which one of the following expresses the general principle that underlies the above argument?

- A. Labour is labour and should have the same value whoever is performing it.
- B. Public safety issues must always be given priority over economic advantages.
- C. Rehabilitation should be for the prisoner's benefit, not the employer's.
- D. Prisoners who are given work should not take jobs away from other workers.
- E. Maintenance of public services should be in the hands of skilled professionals only.

Ans: A

30. Recently, packets of Amblers crisps have had 'money-off' coupons inside them. Some coupons are worth 9p, some are worth 14p and some are worth 20p.

George has been collecting these coupons. He has more 14p coupons than 9p coupons, and more 9p coupons than 20p coupons. The total value of all of his coupons is exactly £1.50.

How many coupons has George collected?

- A. 9
- B. 10
- C. 11
- D. 12
- E. 13

Ans: C